

Baryon multiplets in $SU(3)_F$ (u, d, s)

$B = q_1 q_2 q_3 \Rightarrow$ 27 combinations of flavor

$$S = 1/2, 3/2$$

Using $SU(3)_F \Rightarrow 3 \times 3 \times 3 = 1_A + 8_{M_{12}} + 8_{M_{23}} + 10_S$ multiplets

$$\psi_{Bary} = \psi_{space} \psi_{spin} \psi_{Flav} \psi_{color}$$

S
—————
A

$\hookrightarrow$ must be symmetric

$$\frac{100_F}{\psi_{Flav} \equiv \text{symm}} \Rightarrow \psi_{spin} \equiv \text{symm}$$

$$S = 1/2 \quad \uparrow\uparrow\downarrow \quad \uparrow\downarrow\uparrow \quad \downarrow\uparrow\uparrow$$

$\Rightarrow S = 1/2$ cannot exist

$$S = 3/2 \Rightarrow \psi_{spin} \equiv \text{symm} \Rightarrow 10_F \text{ has } S = 3/2$$

$$\frac{1\bar{1}_F}{\text{antisymm. in flavor}}$$

antisymm. in flavor

$$\psi_{1\bar{1}_F} = \psi_{space} \psi_{spin} \psi_{Flav} \psi_{color}$$

S

A
A

$\Rightarrow$ must have ψ_{spin} Antisymm. under any exchange

$$1 \leftrightarrow 2, 2 \leftrightarrow 3, 1 \leftrightarrow 3$$

$S = 3/2$ always symm

$S = 1/2$ cannot be always anti-symm

$\Rightarrow 1\bar{1}_F$ cannot exist

8F

$$8_{F,12} + 8_{F,23}$$

$$\Psi_8 = \psi_{\text{space}} \underbrace{\psi_{\text{spin}} \psi_{\text{flavor}}}_{A} \psi_{\text{color}}$$

$$\psi_{\text{spin}}^{1/2} \psi_{8,12}$$

Symm.
under exchange
of 1 ↔ 2

both spin and

flavor

$$\psi_{\text{spin}}^{1/2} \psi_{8,123}$$

Symm.

under 2 ↔ 3

Flavor + spin.

$$\equiv \psi_{S_{12}}^{1/2} \psi_{F_{123}}$$

$SU(3)_F$ used for classification of hadrons

$\Rightarrow SU(3)_C$ for color of quarks.

$\psi_{\text{color}} \equiv A \Rightarrow$ particle is color singlet

Hypothesis: all physical particles are color singlets.
colorless.

Baryons: color singlet $R + G + B = 0$

$$3_C \times 3_C \times 3_C = 1_C + 8_C + 8_C + 10_C$$

color singlet for baryons.

Mesons: $q_i \bar{q}_j$

$$3_C \times \bar{3}_C = 1 + 8$$

color singlet / colorless.

$$R + \bar{R} = 0$$

$$B + \bar{B} = 0$$

$$G + \bar{G} = 0$$

SU(3) color, QCD

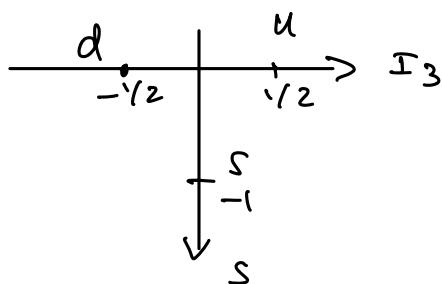
SU(3)_F

I_3 , Strangeness (diagonal)

$$Q = I_3 + \frac{B+S+\cancel{-}}{2} = I_3 + \frac{Y}{2}$$

Y : Hypercharge

Flavor Space



$$B = 1/3$$

$$Y = B + S$$

$$Y_u = \frac{1}{3} + 0 = 1/3$$

$$Q_s = 0 + \frac{1}{2} \left(+\frac{1}{3} - 1 \right) = \frac{1}{2} \left[-\frac{2}{3} \right] = -1/3$$

$$Y_s = B + S = \frac{1}{3} - 1 = -\frac{2}{3}$$

SU(3): 8 generators.

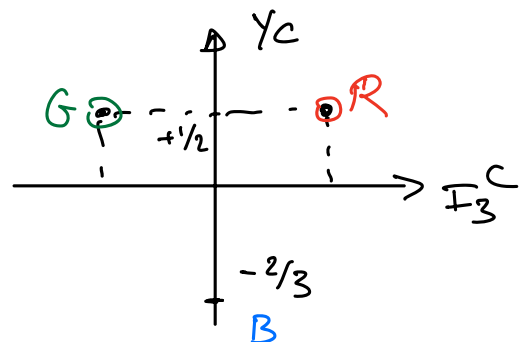
2 diagonal generators

SU(3) color

2 diagonal operators

I_3^C : color isospin

Y_C : color hypercharge



$$R = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \quad G = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \quad B = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

QCD: Quantum Chromodynamics

Yang-Mills theory of interaction of colors.

8 generators $\rightarrow$ 8 gluons mediators of QCD.

$$\text{gluons: } 3_C \times \bar{3}_C = \underbrace{1}_{\text{color singlet}} + 8 \text{ color octet}$$

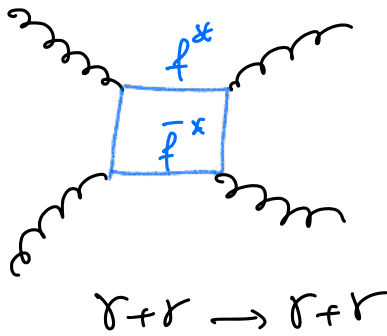
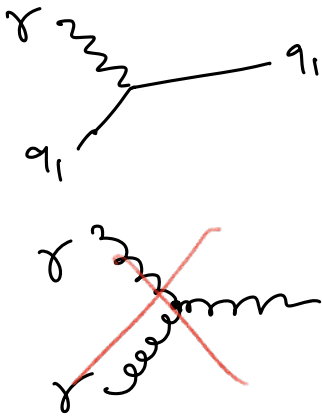
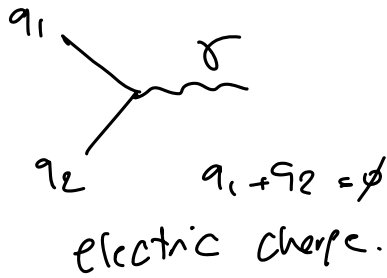
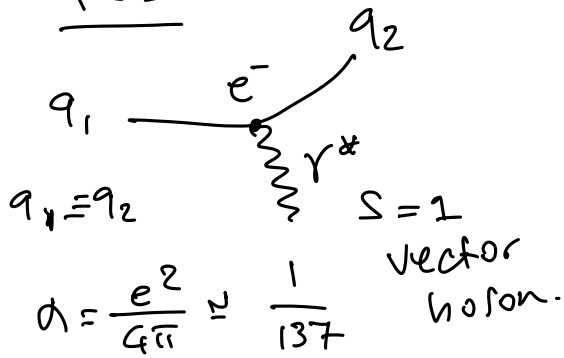
$\Rightarrow$ only 8 colored gluons.

1 colorless gluon does not exist

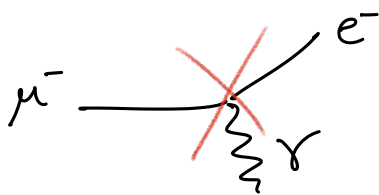
SU(3)_C: colored objects are not free states available in nature.

color confinement

QED

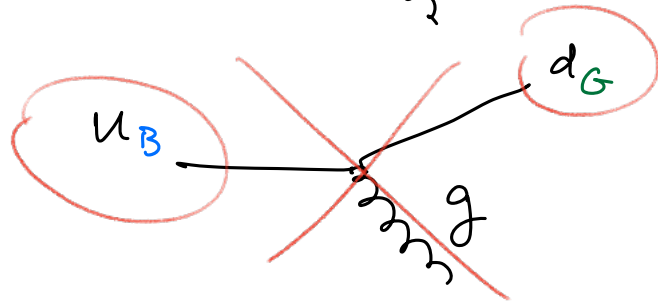
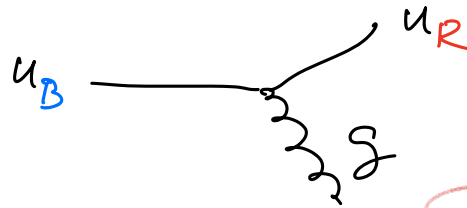
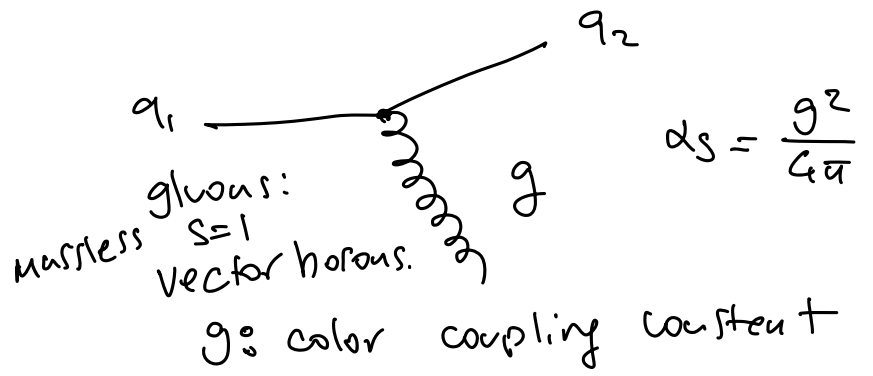


$f = e, \mu, \tau, u, d, s, c, b, t$

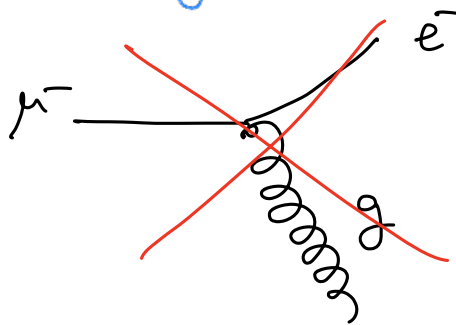


lepton Flavor violation.
not observed yet

QCD

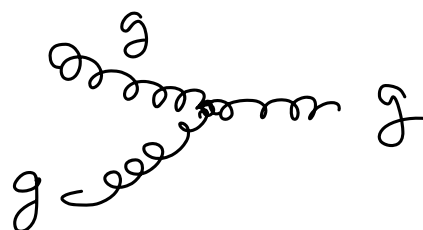


gluons do not change flavor of quarks.
only color



leptons do not have color charge

Gluons carry color charge



Gluons

$$3_C \times \bar{3}_C = 11 + 8$$

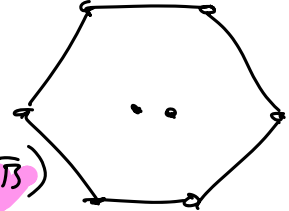
8: colored gluons

1: colorless gluon $\Rightarrow$ no evidence of free colorless gluon.

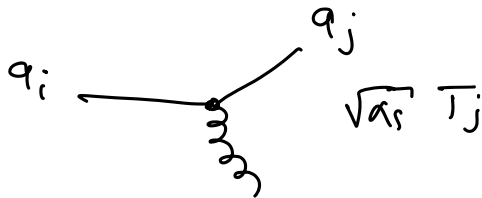
color singlet $\frac{1}{\sqrt{3}} (R\bar{R} + B\bar{B} + G\bar{G})$

gluon octet: $R\bar{G}, R\bar{B}, G\bar{R}, G\bar{B}, B\bar{R}, B\bar{G}$

$$\frac{1}{\sqrt{2}} (R\bar{R} - G\bar{G}), \frac{1}{\sqrt{6}} (R\bar{R} + G\bar{G} - 2B\bar{B})$$



gluons: carry both color and anti-color



$$\sqrt{\alpha_s} T_j$$

$j=1, \dots, 8$ gluons.

T_j : generators of $SU(3)$ color

$$R = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \quad G = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \quad B = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

T_j : Cartan-Weyl matrices known since early 1900

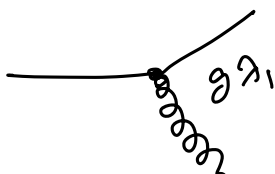
$$T_j = \frac{1}{2} \lambda_j \quad \text{Gell-Mann matrices.}$$

$$\lambda_1 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}; \quad \lambda_2 = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}; \quad \lambda_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\lambda_4 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}; \quad \lambda_5 = \begin{pmatrix} 0 & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & 0 \end{pmatrix}; \quad \lambda_6 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$\lambda_7 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix}; \quad \lambda_8 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix} \cdot \frac{1}{\sqrt{3}} (R\bar{R} + G\bar{G} - 2B\bar{B})$$

λ_1 :

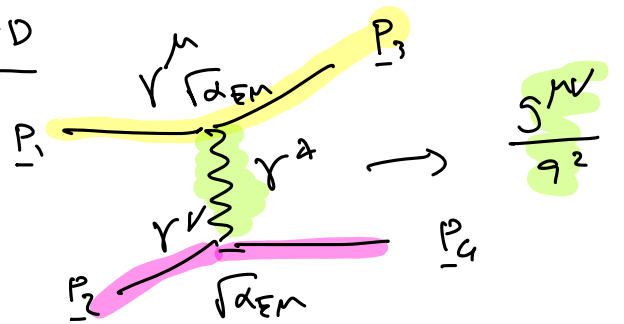


$$q_1 = \frac{100}{\bar{R}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} G$$

λ_1

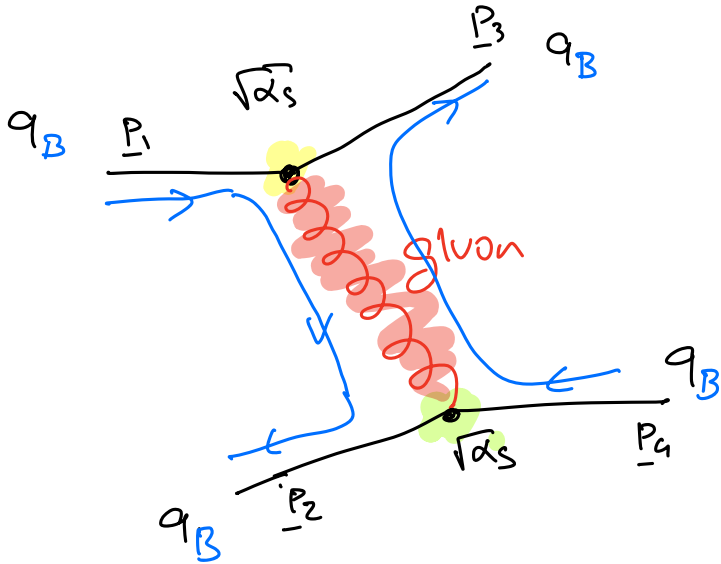
$$q_B + q_B \longrightarrow q_B + q_B$$

QED



$$q = p_1 - p_3$$

$$M_{QED} \sim (\sqrt{\alpha})^2 \bar{u}(p_3) \gamma^\mu u(p_1) \Pi(q) \bar{u}(p_4) \gamma^\nu u(p_2)$$



QCD

$$M_{QCD} \sim (\sqrt{\alpha_s})^2 \bar{u}_B(p_3) \gamma^\mu \frac{\lambda_8}{2} u_B(p_1) \Pi(q^2) \bar{u}_B(p_4) \gamma^\nu \frac{\lambda_8}{2} u_B(p_2) \times M_{QED}$$

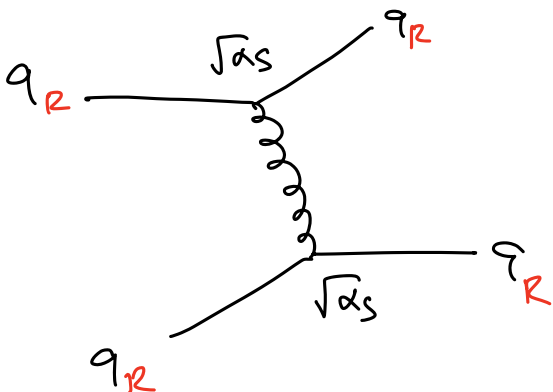
$$\sqrt{\alpha_s} \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & -2 \end{pmatrix} \frac{1}{2} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix} \frac{1}{\sqrt{3}} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \sqrt{\alpha_s} \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & -2 \end{pmatrix} \frac{1}{2} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix} \frac{1}{\sqrt{3}} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$M(BB \rightarrow BB) \sim (\sqrt{\alpha_s})^2 \left(\frac{-2}{2\sqrt{3}} \right) \left(\frac{-2}{2\sqrt{3}} \right) = \frac{\alpha_s}{3}$$

color factor from QCD

$$q_R + q_R \longrightarrow q_R + q_R$$

$$R = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$



$$\begin{pmatrix} 1 & 0 & 0 \end{pmatrix} (\lambda_3 + \lambda_8) \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$M_{QCD}(RR \rightarrow RR) = M_3 + M_8$$

$$\begin{array}{c} R \\ | \\ R - \text{---} \text{---} R \\ | \\ R \end{array} = \begin{array}{c} R \\ | \\ R - \text{---} \text{---} R \\ | \quad \lambda_3 \\ R \end{array} + \begin{array}{c} R \\ | \\ R - \text{---} \text{---} R \\ | \quad \lambda_8 \\ R \end{array}$$

$$M_3 \sim \frac{\sqrt{\alpha_s}}{2} \overline{100} \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \frac{\sqrt{\alpha_s}}{2} \overline{100} (\lambda_3) \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \left(\frac{\sqrt{\alpha_s}}{2} \right)^2 = \left(\frac{1}{2} \right)^2 \alpha_s$$

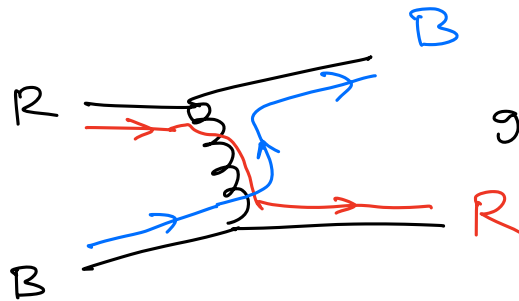
$$\begin{aligned}
 M_8 &\sim \frac{\sqrt{\alpha_s}}{2} \overline{100} \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \frac{\sqrt{\alpha_s}}{2} \overline{100} \lambda_8 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \\
 &= \left(\frac{\sqrt{\alpha_s}}{2} \right)^2 \left(\frac{1}{\sqrt{3}} \right)^2 = \frac{1}{3} \frac{\alpha_s}{4}
 \end{aligned}$$

$$M(RR \rightarrow RR) = \frac{\alpha_s}{4} + \frac{1}{3} \frac{\alpha_s}{4} = \left(1 + \frac{1}{3} \right) \frac{\alpha_s}{4} = \frac{\alpha_s}{3}$$

$$M(BB \rightarrow BB) = \frac{\alpha_s}{3}$$

Some value but from different gluons

$$q_R + q_B \rightarrow ?$$



gluon $R\bar{B}$
which λ_i ?

$$q_G + \bar{q}_G \rightarrow q_i + \bar{q}_i$$

$i = R, B, G$

